

Route Tables

The Forwarding Logic That Steers Every VPC Packet

Interview Preparation Guide

Problem & motivation - longest-prefix match - targets & propagation - packet flow - failure modes
- scaling - security - observability & automation - cost - troubleshooting - real-world patterns

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Executive Summary

A route table is the control surface that decides the next hop for every packet leaving a subnet. It is an ordered list of routes, each mapping a destination CIDR (or managed prefix list) to a target such as the local VPC, an Internet Gateway, a NAT Gateway, a peering connection, a Transit Gateway, or a gateway endpoint. The router uses longest-prefix match to choose the most specific route. Route tables are where 'public vs private' is actually decided, and where most VPC connectivity bugs are found. This guide covers internals, packet flow, failure modes, scaling, security, observability, automation, cost, and troubleshooting.

1. The Problem It Solves

Resources need a deterministic, auditable way to decide where traffic goes: stay inside the VPC, exit to the internet, reach a peer or on-prem network, or hit a managed service privately. Route tables provide that decision layer per subnet, decoupled from the instances themselves.

Core needs route tables address

- **Next-hop decisions:** Map destinations to gateways/targets without touching instance OS routing.
- **Segmentation:** Different subnets get different egress (public vs private vs isolated) by associating different tables.
- **Connectivity composition:** Combine IGW, NAT, peering, TGW, VGW, and endpoints into one coherent forwarding policy.
- **Auditability:** Routing becomes declarative configuration, reviewable and versioned.

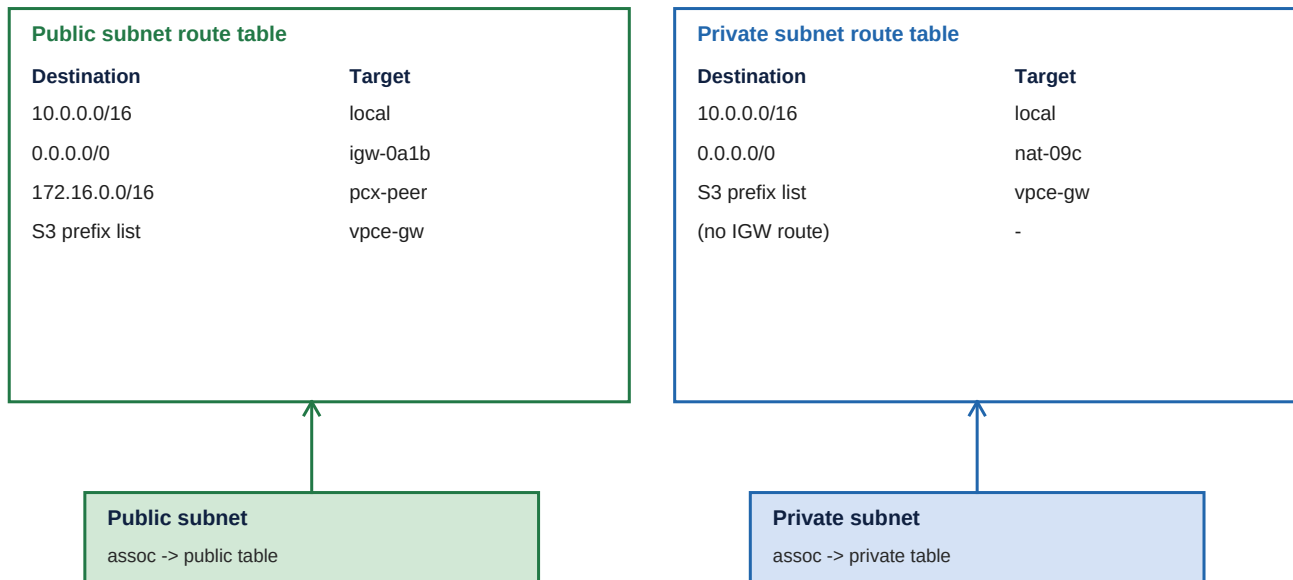
2. How It Works Internally

Every VPC has an implicit, highly available router. Route tables are the rules that router consults. Each table contains an un-removable local route for the VPC CIDR (and each secondary CIDR) plus the routes you add. Routes can be static (you set them) or propagated (learned dynamically from a VGW/Transit Gateway via BGP).

- Selection uses longest-prefix match; for equal prefixes, static routes are preferred over propagated ones.
- Targets include: local, igw, nat, eni, pcx (peering), tgw, vgw, gateway endpoints, and egress-only IGW (IPv6).
- Managed prefix lists let you reference a named set of CIDRs (e.g., S3) in a route or rule.
- A subnet uses its explicitly associated table, else the VPC main table.

Reference Diagram - How a route table steers traffic

A route table is an ordered set of (destination CIDR -> target) rules. Each subnet is associated with one route table; the router applies longest-prefix match to pick the next hop. Below: a custom 'public' table and the rules a private subnet would use instead.



Longest-prefix match wins: 10.0.5.7 matches local (/16) over 0.0.0.0/0. The local route is implicit and cannot be removed. A subnet not explicitly associated falls back to the VPC 'main' route table.

3. Traffic Flow, Packet Flow & Routing Decisions

Decision order for an outbound packet

- Packet leaves ENI; security group egress evaluated (stateful).
- Router scans the subnet's route table for the most specific matching destination (longest-prefix match).
- Local route always wins for in-VPC destinations and cannot be overridden.
- Static vs propagated tie-break: static preferred; among propagated, BGP attributes decide.
- Chosen target forwards the packet (IGW, NAT, TGW, peering, endpoint, ...).

Common gotcha - asymmetric routing

If the outbound subnet routes via NAT but the return-path subnet/table routes differently (or a peering route is missing on one side), you get one-way connectivity. Routes must be consistent in BOTH directions.

4. Advantages & Disadvantages

Advantages

- Declarative, per-subnet control of all forwarding.
- Composes many connectivity types into one policy.
- Dynamic propagation (BGP) reduces manual route sprawl with VGW/TGW.
- Easy to reason about with longest-prefix match.

Disadvantages / costs

- Route-entry limits cap how many explicit routes a table can hold.
- Easy to misconfigure (wrong target, missing return route, overlaps).
- No native ordering/priority beyond prefix length and static/propagated.
- Propagated + static interactions can surprise operators.

5. Common Failure Modes

Failure	Symptom	Typical root cause
Missing 0/0 route	No internet egress	Forgot IGW/NAT default route on the subnet's table.
Wrong association	Subnet unexpectedly public/private	Subnet bound to main table instead of intended one.
Asymmetric routes	One-way / intermittent	Return path lacks a matching route (peering/TGW/NAT).
Blackhole route	Silent drops	Target deleted; route now points to nothing (blackhole).
Overlapping prefixes	Wrong next hop chosen	A more specific route steers traffic unexpectedly.
Entry-limit hit	Cannot add route	Too many explicit CIDRs; needs summarization.

6. Scaling Considerations

- **Summarize:** Aggregate CIDRs (supernet) to stay under route-entry limits.
- **Prefer dynamic:** Use BGP propagation with VGW/TGW so routes scale without manual entries.
- **Prefix lists:** Reference managed prefix lists to keep rules compact and centrally updatable.
- **Hub routing:** At many-VPC scale, centralize with Transit Gateway route tables instead of N^2 peering routes.
- **Separation:** Use distinct tables per tier to keep each small and clear.

7. Security Implications

- Routing IS a security control: removing a 0/0->IGW route makes a subnet private.
- Avoid overly broad routes that expose isolated/data subnets.
- Use isolated tables (only local + endpoints) for sensitive data tiers.
- Alert on route changes - they can silently open or sever connectivity.
- Centralized inspection: route egress through a firewall/appliance subnet.

8. Observability, Reliability & Automation

Observability

- Reachability Analyzer to prove a path exists (or find the broken hop).
- Flow Logs to see whether traffic reaches the expected next hop.
- Detect blackhole routes (deleted targets) via config monitoring.

Reliability

- The VPC router itself is HA and managed; design redundant targets per AZ.
- Ensure failover paths exist (e.g., per-AZ NAT) so one target loss is contained.

Automation

- Define routes in IaC; avoid console drift.
- Policy-as-code to forbid 0/0->IGW on data-tier tables.

9. Cost Considerations

Cost driver	Why it adds up	Mitigation
Route tables themselves	Free - no charge for tables or routes.	N/A; the cost is in the targets they point to.
NAT/TGW targets	Routes to NAT/TGW drive data-processing cost.	Use gateway endpoints (free for S3/DynamoDB) where possible.
Suboptimal paths	Routing cross-AZ/cross-region adds transfer cost.	Design routes to keep traffic local when feasible.

10. Troubleshooting Approach

When traffic does not flow, suspect routing first:

1. Which table is the source subnet associated with?
2. Is there a matching route to the destination, and what is the target?
3. Is the target healthy (not a blackhole / deleted)?
4. Does the RETURN subnet/table have a matching route back?
5. Confirm with Reachability Analyzer + Flow Logs.

Mnemonic

'Association -> Match -> Target -> Return.' Verify both directions; asymmetric routing is the classic silent failure.

11. Real-World Examples & Patterns

- **Public/private split:** Separate tables: public->IGW, private->NAT, both with gateway endpoints.
- **Inspection VPC:** Routes steer all egress through a firewall appliance subnet before the internet.
- **Hub-and-spoke:** TGW route tables segment which VPCs can reach which, centrally.
- **Hybrid:** VGW route propagation injects on-prem prefixes learned via BGP.

12. Senior-Level Trade-offs & Talking Points

- Static vs propagated routes: predictability vs scalability/auto-failover.
- Many specific routes vs summarized routes: precision vs entry limits.
- Per-subnet tables vs shared tables: isolation vs management overhead.
- Distributed routing (peering) vs centralized (TGW route tables): latency/cost vs operability.

Whiteboard checklist

(1) Draw subnets and their associated tables. (2) List destination->target rules. (3) Show longest-prefix match picking a hop. (4) Trace return path. (5) Discuss propagation, blackholes, limits, and inspection routing.

13. Quick Glossary

Term	Meaning
Local route	Implicit, un-removable route for the VPC CIDR(s).
Longest-prefix match	Most specific matching CIDR wins.
Target	Next hop: igw, nat, pcx, tgw, vgw, eni, endpoint, etc.
Propagated route	Route learned dynamically via BGP from VGW/TGW.
Main route table	Default table used by subnets with no explicit assoc.
Blackhole	Route whose target no longer exists; traffic is dropped.
Prefix list	Named, reusable set of CIDRs for routes/rules.